The Development History of Human Exploration of Outer Space

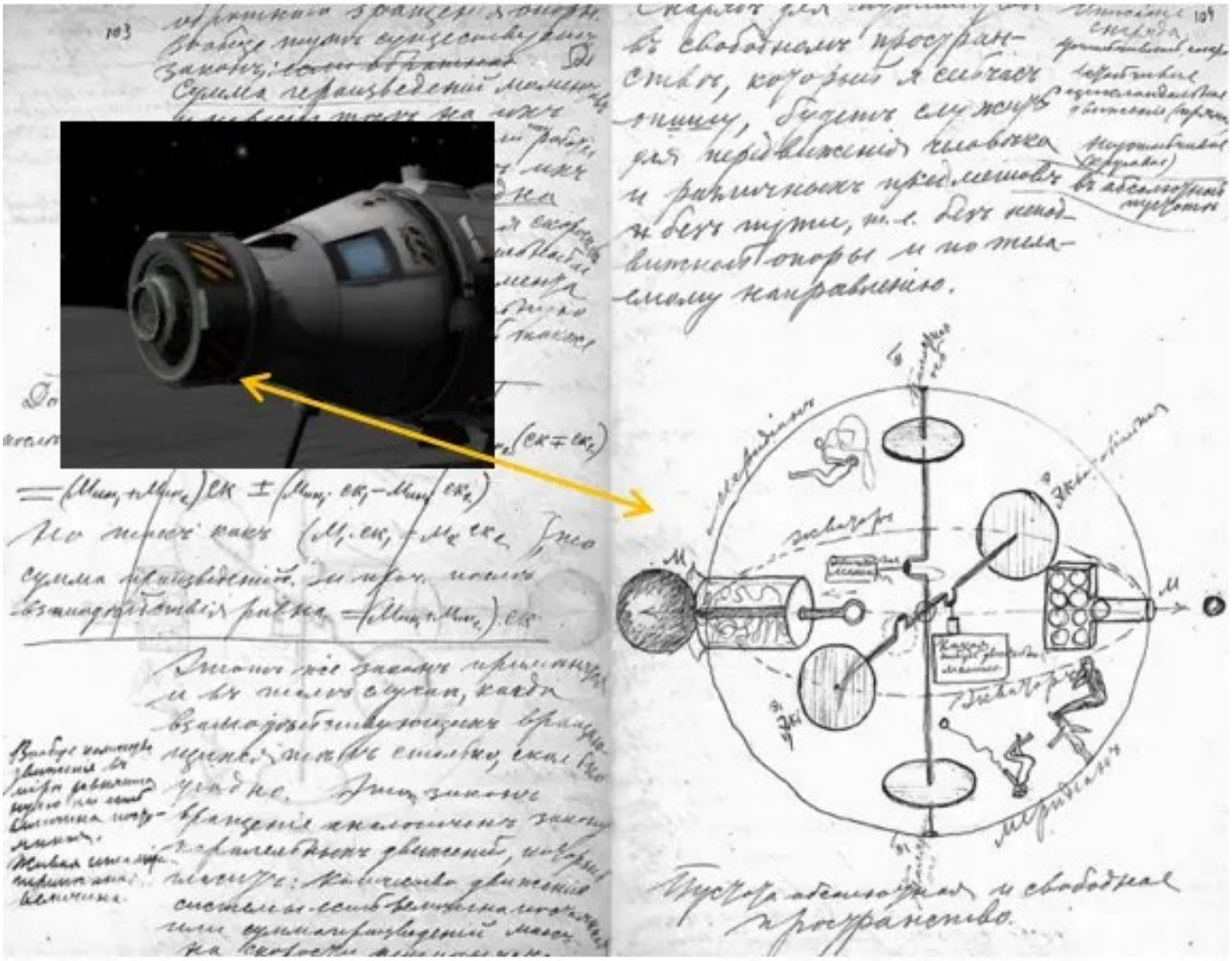
Human exploration of outer space is a series of scientific practices driven by human curiosity about the unknown universe, the desire to break free from the constraints of Earth's gravity, and the need to expand the boundaries of survival. This journey, spanning nearly a century, can be roughly divided into four core stages: theoretical foundation, the space race during the Cold War, the ebb of exploration enthusiasm, and the rise of commercial spaceflight. The process of exploration is directly influenced by the historical background, international pattern, technological level, and economic strength, showing alternating characteristics of phased breakthroughs and stagnation. In the form of an expository text, this article objectively sorts out the development context of human exploration of outer space, clarifies the core events, technological breakthroughs, and historical impacts of each stage, balances the completeness and rigor of the content, responds to core requirements, avoids literary lyricism and excessive colloquialism, and clearly presents the connection between humans and outer space.



The theoretical foundation of human space exploration was laid by the Russian scientist Konstantin Tsiolkovsky. In the early 20th century, human understanding of outer space was still in its initial stage, confined to imagination and sporadic experiments, lacking systematic theoretical support. Rocket technology was only applied to small military explosive devices and had no direct connection with space exploration; most people believed that manned spaceflight was not feasible. As a middle school mathematics teacher, Tsiolkovsky, driven by his own interest in space exploration, conducted research in his spare time without professional research conditions, research team support, or official funding. Through manual calculations, simple model deductions, and astronomical observations, he gradually constructed a theoretical system of rocket motion.

In 1903, Tsiolkovsky published the article "Exploration of Cosmic Space by Means of Reaction Devices", in which he systematically elaborated on the basic principles of rocket motion for the first time and derived the Tsiolkovsky Rocket Equation. This equation accurately reveals the quantitative

relationship between rocket flight speed, fuel mass, and the rocket's own mass. Its core conclusion is: no matter how much the fuel efficiency is optimized, a single-stage rocket cannot reach the first cosmic velocity (7.9 kilometers per second) required to escape Earth's gravity; only by adopting a multi-stage rocket design, discarding the fuel-exhausted stages step by step to reduce the rocket's own weight, can the flight speed be gradually increased, and finally break free from Earth's gravity to enter outer space. This theoretical breakthrough broke the cognitive barrier of human space exploration and transformed space exploration from an illusion into a scientific goal with a clear path.



In addition to the rocket equation, Tsiolkovsky also put forward a series of forward-looking concepts, including the specific design ideas of multi-stage rockets, the concept of space station construction, the technical feasibility of interstellar navigation, and the possibility of human landing on the moon and Mars. Later, he improved his theoretical system through many papers, further putting forward the initial idea of space migration and space colonization, believing that humans will inevitably break through the boundaries of the Earth and find new living spaces in the universe. Although these concepts were not widely recognized at that time due to being too advanced, they provided a clear direction for the development of aerospace technology in the following century.

Tsiolkovsky's theory was not immediately transformed into practical aerospace practice. The core reason was the insufficient technological level and industrial capacity at that time, the lack of fuel capable of carrying large thrust, high-strength rocket body materials, and precise control

technology. Moreover, the attention of various countries was focused on ground development and war preparation, and they were unwilling to invest a lot of funds in researching space-related technologies with high risks and long return cycles. It was not until the 1920s and 1930s that with the progress of industrial technology and the attention of various countries to military technology, his theory was gradually concerned by scientists from Germany, the United States, the Soviet Union and other countries, and became the core theoretical basis for aerospace technology research and development.

During this period, the German Wernher von Braun team took the lead in transforming the theory into practice. During World War II, commissioned by the German military, the team developed the V-2 rocket for long-range military strikes. Absorbing Tsiolkovsky's rocket theory and combining with the industrial technology at that time, they successfully developed the first rocket in human history that could break through the atmosphere. The V-2 rocket was about 14 meters long, 1.65 meters in diameter, and about 13 tons in weight. It was propelled by liquid fuel, with a maximum flight speed of 2.9 kilometers per second and a maximum range of about 320 kilometers. Although its purpose was military strikes, which caused great damage to Britain and other countries, it objectively verified Tsiolkovsky's theory, accumulated key technologies such as rocket propulsion, rocket body design, and fuel storage, and laid the foundation for the rapid development of aerospace technology after the war.

After World War II, the United States and the Soviet Union became the two major superpowers in the world, and the Cold War officially began. The two countries launched comprehensive confrontations in politics, military, science and technology, and the space field became the core battlefield of competition between the two sides. The achievements of space exploration were not only a reflection of scientific and technological strength, but also a symbol of international discourse power and military deterrence. This political demand gave birth to a fierce space race, and also provided huge financial support and sufficient human resources for space exploration, promoting human space exploration into a period of rapid development.

With the institutional advantage of concentrating efforts on major tasks, the Soviet Union took the lead in making breakthroughs in the space race and achieved many "firsts in human history". On October 4, 1957, the Soviet Union successfully launched Sputnik 1, the first artificial Earth satellite in human history. The satellite was 58 centimeters in diameter, about 83.6 kilograms in weight, and carried a simple radio transmitting device that could send weak radio signals to the Earth. This event marked the official entry of humans into the space age. It not only demonstrated the Soviet Union's aerospace strength, but also stimulated the United States to accelerate the pace of aerospace research and development, pushing the space race into a white-hot stage.

On April 12, 1961, Soviet astronaut Yuri Gagarin boarded the Vostok 1 manned spacecraft and orbited the Earth for 1 hour and 48 minutes before returning to the ground safely, becoming the first human to enter outer space. This achievement proved that humans can survive in the space environment and laid the foundation for the development of manned spaceflight technology. Since then, the Soviet Union continued to promote the aerospace industry. In 1963, Valentina Tereshkova became the first female astronaut to enter outer space; in 1965, Alexei Leonov completed the first spacewalk in human history, expanding the scope of human activities in space; in 1971, the Soviet Union launched Salyut 1, the first space station in human history, opening the way for long-term human stay in space.

Faced with the Soviet Union's leading advantage, the United States launched the Apollo program in May 1961. At that time, President Kennedy announced that the United States would land humans on the moon within 10 years. The core goal was to reverse the disadvantage in the space race and demonstrate its own scientific and technological strength and comprehensive national strength. This

program was one of the largest, most costly, and most technologically complex aerospace projects in human history. It invested a total of about 25 billion US dollars (equivalent to more than 200 billion US dollars today), employed more than 400,000 people, involved more than 20,000 enterprises, more than 120 universities and research institutions, covering rocket research and development, spacecraft design, life support, orbit calculation and other fields.

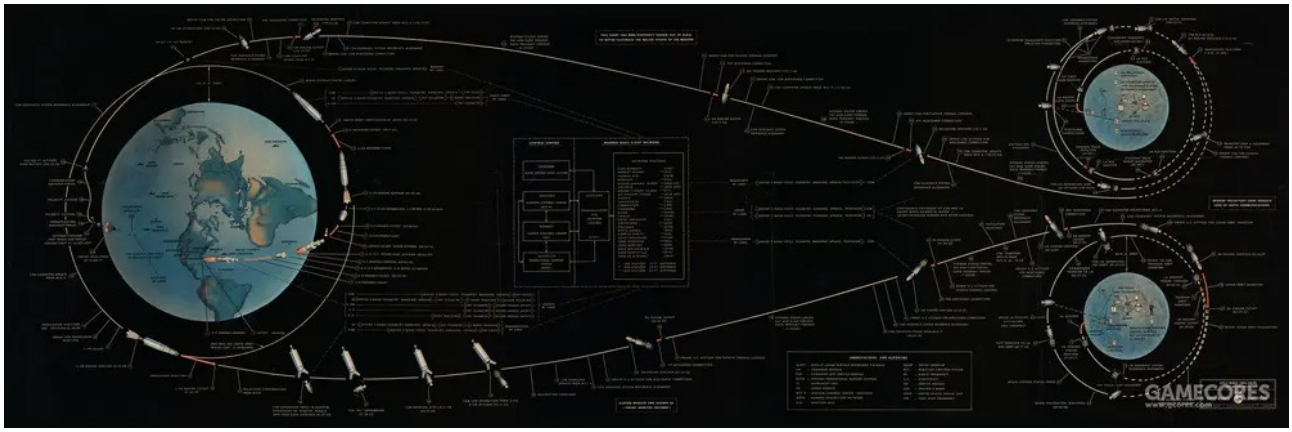


The most iconic significance of the Apollo moon landing is that scientists completed the seemingly impossible task through rigorous calculations and repeated experiments under the background of extremely backward computer technology at that time. In the 1960s, computer technology was in its primary stage, with large volume, slow computing speed, and small storage capacity. There was no graphical interface or convenient programming tools. All calculations and data processing relied on engineers manually writing machine language code, paper-and-pencil calculations, and brain deductions. Taking the guidance computer used by Apollo 11 as an example, it had only 64KB of memory and a main frequency of 1.024MHz, with a computing speed of about 100,000 times per second. Compared with the current ordinary smartphones (with computing speed of billions of times per second and memory of several GB), the gap is tens of thousands of times.

Under the backward technical conditions, scientists faced three core difficulties. First, the accuracy of orbit calculation. The distance between the Earth and the moon is about 380,000 kilometers. The round-trip of the spacecraft requires precise calculation of orbit parameters. A small error can cause the spacecraft to deviate from the orbit and eventually crash. Due to the lack of advanced simulation technology, scientists could only rely on manual calculations, repeated deductions combined with astronomical observation data, and even used dozens of computers for parallel computing to ensure the accuracy of parameters. Second, spacecraft control and navigation. In space, the spacecraft cannot rely on ground navigation, but only on its own guidance computer and manual operation by astronauts. However, the computer cannot handle complex emergencies, so astronauts need to have strong emergency response capabilities and can manually control the spacecraft when the computer fails. Third, the research and development of life support systems. There is no air, no gravity, extreme temperature, and strong radiation in space. Scientists developed systems for recycling air, water, and food through repeated experiments, and designed spacesuits that can resist space radiation and adapt to weightless environments to ensure the safety of astronauts.



During the implementation of the Apollo program, there was a major setback. In 1967, the Apollo 1 spacecraft caught fire during a ground test, killing 3 astronauts. After the accident, scientists comprehensively analyzed the causes, improved the spacecraft design, improved safety measures, and gradually promoted the implementation of the program. On July 20, 1969, the Apollo 11 spacecraft carried three astronauts, Neil Armstrong, Buzz Aldrin, and Michael Collins, to the lunar orbit. Armstrong and Aldrin boarded the lunar module and successfully landed on the lunar surface. Armstrong became the first human to set foot on the moon. The two spent about 2 hours and 30 minutes on the lunar surface, collecting rock samples, installing experimental equipment, recording environmental data, and then returned to the lunar module, rendezvoused with Collins in orbit, and successfully returned to the Earth.



By 1972, the Apollo program had carried out 17 missions, including 6 successful manned moon landings and 11 test or circumlunar missions. A total of 12 astronauts were sent to the moon, and about 382 kilograms of lunar rock and soil samples were brought back, providing valuable data for human research on the origin of the moon and the evolution of the universe. At the same time, the

program promoted the all-round progress of aerospace technology, accumulated a lot of technical experience in rocket propulsion, spacecraft control, life support and other fields. These technologies were later gradually applied to civilian fields, promoting the overall development of global science and technology.

After the end of the Cold War, human space exploration entered a period of ebb. The dissolution of the Soviet Union in 1991 made the US-Soviet space race lose its political motivation, and both countries greatly reduced their investment in the aerospace field. The proportion of NASA's budget in the federal budget dropped from 4.5% at its peak to less than 0.5%, the manned moon landing program was completely shelved, and deep space exploration fell into a stalemate. The aerospace technologies and facilities left by the Soviet Union were inherited by Russia. Due to economic constraints, the pace of aerospace development slowed down, only maintaining basic manned spaceflight and space station operations.

There are three core reasons for the ebb of space exploration: first, the high cost. Space exploration requires huge capital investment and has a long return cycle. After the Cold War, various countries paid more attention to ground people's livelihood development, prioritizing funds to education, medical care, economic construction and other fields, resulting in a significant reduction in investment in aerospace. Second, technical bottlenecks. The rapid breakthroughs during the Cold War mainly relied on politically driven concentrated investment. There were no major breakthroughs in basic science, and the development of aerospace technology fell into a bottleneck, making it difficult to achieve new leapfrog progress. Third, the decline of public enthusiasm. The space race during the Cold War attracted the attention of the global public, but after the end of the Cold War, space exploration lacked clear competitive goals, and public attention gradually declined, further affecting the willingness of various countries to invest in the aerospace field.

During this period, human space exploration did not completely stop, but entered an accumulation stage. Countries gradually turned to cooperative exploration. In 1998, the International Space Station officially started construction, jointly participated by 16 countries including the United States, Russia, the European Union, Japan, and Canada. It became the core platform for long-term human stay in space and scientific experiments, promoting the coordinated development of aerospace technology. At the same time, countries began to focus on practical fields such as near-Earth orbit utilization and satellite applications, gradually expanding the application value of space exploration and laying the foundation for subsequent recovery.



In the early 21st century, the rise of commercial spaceflight promoted human space exploration into a recovery stage, among which SpaceX, as a new space force, became the core driving force. Founded by Elon Musk in 2002, SpaceX's core goal is to reduce the cost of space exploration, realize human Mars migration, and make humans a "multi-planetary species". Different from the government-led aerospace projects during the Cold War, SpaceX is driven by commercial logic, breaking the long-term government monopoly in the aerospace field, and greatly reducing the cost of space launch through technological innovation and business model optimization.



SpaceX's core technological breakthrough is the reusable rocket technology. Previously, traditional rockets were mostly used once, and the rocket body crashed after launch, leading to high launch costs; while SpaceX's Falcon 9 rocket achieved orbital-level vertical landing and reusability, which can recover and reuse the first stage of the rocket, greatly reducing the launch cost - the cost of a single launch of a traditional rocket is more than 100 million US dollars, while the cost of a single launch of the Falcon 9 after reuse can be reduced to tens of millions of US dollars or even lower.

From the perspective of development history, in 2008, SpaceX's Falcon 1 rocket successfully achieved the first orbital launch by a private company, marking the entry of commercial spaceflight into a substantive development stage; in 2015, the Falcon 9 rocket achieved the first vertical landing, breaking through the core technology of reusability; in 2020, the Crew Dragon spacecraft carried astronauts to the International Space Station, becoming the first private company to realize manned spaceflight, breaking the government's monopoly on manned spaceflight; in 2022, the Starship prototype completed its first orbital test flight, laying the foundation for subsequent Mars exploration. Up to now, SpaceX has become a leading enterprise in the global commercial

spaceflight field. Its Starlink program plans to deploy tens of thousands of satellites to build a global satellite Internet, further expanding the application value of space.



The rise of SpaceX has not only activated the global commercial spaceflight market, but also promoted the recovery of the aerospace industry in various countries. NASA has carried out in-depth cooperation with SpaceX, entrusting it with tasks such as manned transportation and cargo supply to reduce its own investment; other countries have also increased their investment in the aerospace field, promoting the coordinated development of commercial spaceflight and national spaceflight. At the same time, SpaceX's technological innovation has also driven the progress of global aerospace technology. Reusable rockets, Mars exploration technology, etc., have become the focus of aerospace research and development in various countries.

Looking back at the nearly 100-year history of human exploration of outer space, from Tsiolkovsky's theoretical foundation, to the technological explosion during the Cold War, and then to the ebb and recovery in the post-Cold War era, human exploration of outer space has never stopped. In this process, humans have broken through one technical difficulty after another, achieving leaps from theory to practice, from unmanned exploration to manned spaceflight, and from government leadership to commercial participation. Space exploration has not only promoted the progress of aerospace technology, but also driven the development of many fields such as basic science, materials science, and electronic technology. At the same time, it has satisfied human curiosity about the unknown universe and expanded the boundaries of human survival.

At present, human space exploration is entering a new stage of development, with commercial spaceflight and national spaceflight advancing in coordination. Mars exploration, space migration, and near-Earth orbit utilization have become core directions. Although space exploration still faces problems such as high cost, difficult technology, and high risk, human curiosity and desire for exploration of outer space will continue to promote humans to break through boundaries and achieve their own development and transcendence in the connection with outer space.